

Reasoning on Sidewalks: Generating Safety-Critical Pedestrian-Delivery Robot Interactions via Causal Diffusion

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Abstract—Delivery robots have been deployed on large-scale citywalks to provide convenient urban services. However, interacting with pedestrians remains a big challenge for delivery robots, especially in corner cases such as falls and sudden appearances in highly occluded sidewalk environments. Due to the limited deployment of delivery robots, data-driven generative models from observational data struggle with the efficiency of sampling corner cases. In addition, sparse data often introduce spurious correlations that are not relevant to collisions, degrading the generalization of models between environments. Therefore, we use domain-adversarial neural networks (DANN) to learn invariant causal features across environments that contribute to collisions and propose a diffusion-based generative framework, causal decision diffuser (CDD). CDD uses causal features together with collision rewards as conditional signals to discover and generate collision-prone scenario parameters rather than searching scenarios based on limited observational data. By learning collision-relevant causal invariant features across diverse environments rather than merely capturing spurious correlations in the data, CDD significantly improves model generalization to unseen, challenging urban delivery scenarios. Compared with existing reinforcement learning (RL) and generative methods, experiments in three classic traffic scenarios show that CDD achieves higher efficiency in generating hazardous scenarios and demonstrates stronger generalization between environments.

I. INTRODUCTION

Micromobility, exemplified by delivery robots, is emerging as a new paradigm for short-distance travel [1]. Unlike autonomous vehicles (AVs) that operate on structured roadways, delivery robots are typically deployed in unstructured environments where they share space with pedestrians. Consequently, they must face pedestrian scenarios that are more complex and less predictable than those of conventional AVs. Existing delivery robot evaluations focus on routine tasks such as navigation [2] and obstacle avoidance [3]. However, when pedestrians exhibit atypical behaviors, the robot’s policy can fail and lead to accidents [4]. Moreover, compared to AVs, the limited deployment results in a insufficient representation of safety-critical scenarios in available observational datasets. For example, a robot can encounter a pedestrian who is alcoholic and falls in its path, a rare event for which the policy does not have prior exposure (Fig. 1a). Such safety-critical scenarios arising from atypical pedestrian behaviors represent a challenge and may constitute the next frontier in evaluating the safety and robustness of delivery robots beyond the standard task.

Generative models trained on observational data are widely used for AVs testing. To address the long tail problem, existing methods often employ importance sampling [5], adversarial training [6], and RL [7] to generate safety-critical

scenarios. However, for delivery robots, such approaches face limitations. Unlike AVs, delivery robots lack large-scale deployment, resulting in limited and biased data that may fail to capture the true distribution of real-world scenarios and introduce spurious correlations irrelevant to collisions. For example, in scenarios where a pedestrian falls, a roadside parked car is irrelevant to the collision, yet the model may mistakenly learn a spurious spatial correlation. In reality, the underlying causal factors, such as the timing of the fall and the robot’s velocity, remain invariant across different environments. Therefore, a robust model should maintain consistent performance regardless of variations in nuisance variables, such as the position of the roadside car. Inspired by invariant learning for cross-domain feature extraction [8] and conditional diffusion models [9], we propose a novel optimization approach. Using causally invariant features as conditional signals to guide a diffusion model to generate safety-critical scenarios with improved cross-environment generalization.

Our key insight is that a collision scenario comprises both causally relevant and irrelevant objects. The collision between a pedestrian and a delivery robot remains invariant to nuisance variables. Motivated by this, we formulate the problem as a conditional generative task, where causal features and collision rewards serve as guiding signals to enable efficient generation of safety-critical scenarios in unseen environments. Therefore, we introduce the causal decision diffuser (CDD), which employs a domain-adversarial neural network (DANN) to extract causal features that trigger collisions across diverse environments. We use a diffusion model to learn the distribution of initial scenario parameters conditioned on causal features and collision rewards. During inference, we apply classifier-free guidance and low-temperature sampling to effectively capture collision-prone scenarios.

The main contribution can be summarized as follows:

- We construct a pedestrian-delivery robot interaction dataset in three corner scenarios (Fig. 1b).
- We propose CDD, a causal generative model that integrates domain invariant learning with conditional diffusion to generate safety-critical scenarios (Fig. 1c).
- We show that CDD improves efficiency and generalization compared to offline RL and generative baselines.

II. RELATED WORK

A. Micromobility and Delivery Robot

Micromobility uses diverse platforms [10], such as delivery robots designed for food and parcel transportation. Due

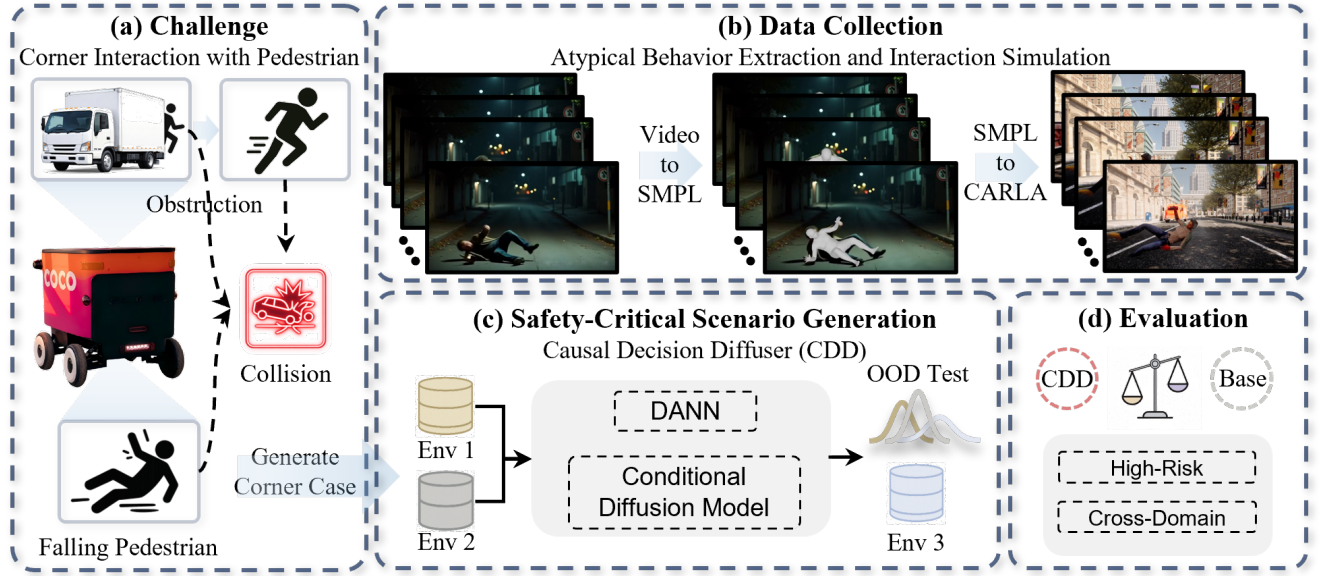


Fig. 1. Overview of the framework. (a) Challenging scenarios involving pedestrians and delivery robots. (b) Extraction of pedestrian atypical behaviors from videos and construction of the interaction dataset. (c) Demonstration of the CDD model: CDD integrates a DANN with a conditional diffusion model to generate cross-environment safety-critical scenarios. (d) Evaluation between CDD model and baseline models.

to their flexibility and lightweight, these robots are well-suited to address the last-mile problem that AVs struggle to solve. Their diverse operational environments require the robot to have robust navigation [2] and obstacle avoidance capabilities [3] to mitigate collisions with pedestrians and cyclists. Existing research focuses on perception [3] and interaction in urban crowded areas [11]. However, these systems often exhibit limited robustness when faced with atypical pedestrian behaviors. Therefore, evaluating performance in safety-critical scenarios is a prerequisite for the reliable deployment of delivery robots.

B. Safety-Critical Scenarios Generation

The generation of safety-critical scenarios is used to evaluate the performance of AVs. Early methods relied on accidents and expert knowledge to define test cases [12]. With the accumulation of driving datasets, data-driven approaches have improved generation efficiency [13]. Importance sampling upweights high-risk but low-frequency scenarios within the natural distribution [5]. Recently, the task has been formulated as a markov decision process (MDP), where RL and adversarial training are employed to perturb the behaviors of critical agents, accelerating the generation of safety-critical events [6]. Among generative models, variational autoencoders (VAE), such as STRIVE [14], generate collisions by perturbing the latent space. Generative adversarial networks (GAN) capture scenario distributions through adversarial optimization [15]. However, GAN often suffers from model collapse, which limits the generation ability. Diffusion models offer superior distribution fitting and controllable conditional inference. For example, Scenario Diffusion [16] imposes constraints during denoising for controllable traffic generation, while DiffScene [17] employs multi-objective adversarial guidance to balance safety, fidelity, and constraints.

However, the above methods mainly learn statistical correlations from observational data. Since collisions are governed by causal chains rather than random coincidences, correlation-based models are prone to performance degradation under out-of-distribution (OOD) shifts. To address this, CausalAF [18] impose causal constraints via mask structures to enhance hazardous scenario generation. Furthermore, CCDiff integrates causal inference into diffusion-based generation to improve controllability [19]. Therefore, unlike purely correlation-based approaches, this work explores causal generative modeling to address cross-environment OOD challenges, enhancing the generalization of safety-critical scenario generation.

C. Causal representation and invariant learning

Enhancing model generalization under OOD shifts is a long-standing challenge. Since observational data hardly cover the full real-world distribution, simply scaling datasets is insufficient. Causal generative models address this by embedding structural causal models (SCMs) into latent spaces, ensuring that latent dependencies reflect true causal structures. Representative works include CausalVAE, which incorporates a learnable causal graph into the VAE model [20], Diff-SCM, which applies diffusion models for counterfactual generation [21], and CausalAF, which uses causal mask to constrain autoregressive generation [18]. However, explicitly defining causal graphs remains challenging and the reliability of the discovered causal relationship is often limited. Invariant causal prediction offers an alternative by identifying causal relationships that remain stable across environments. Methods such as invariant risk minimization (IRM) [22] and CausalDiffRec [23] use multi-environment data to extract cross-domain features, improving robustness against distributional shifts. Therefore, for delivery robot scenarios with

limited data, invariant learning provides a robust mechanism to capture the causal features that trigger collisions, enhancing the generalization of scenario generation models.

III. METHODOLOGY

We formulate the task as a bandit problem, where a conditional diffusion model generates initial scenario parameters with high collision probabilities. As shown in Fig. 2, we employ adversarial training between a DANN and a VAE to extract collision-relevant causal features from datasets spanning the same collision type across diverse environments. The CDD then generates scenarios conditioned on these causal features and collision rewards. During inference, classifier-free guidance and low-temperature sampling are applied to improve the sampling quality. In this way, we establish a causal generative model for safety-critical scenarios.

A. Problem Statement

Decision-making problems are typically addressed via offline RL, which learns an optimal policy from a fixed dataset. However, offline RL is susceptible to a distribution shift, where the inference policy deviates from the training distribution. Addressing this issue requires constraints and extensive hyperparameter tuning. In contrast, the Decision Diffuser [9] avoids these challenges by adopting a maximum-likelihood estimation training.

Hence, we formulate environments as $e \in \mathcal{E} := \{\mathcal{U}, \mathcal{R}, \mathcal{S}, \mathcal{A}\}$. Each environment e defines a consistent distribution over collision-relevant objects and a different distribution on collision-irrelevant objects. A binary label $r \in \mathcal{R} = \{0, 1\}$ indicates the collision states, with $r = 1$ denoting a collision and $r = 0$ a safe interaction. $u \in \mathcal{U}$ denotes the objects in each environment. $s \in \mathcal{S}$ represents the states of objects across 20 consecutive frames following the initiation of the robot's movement:

$$s(u) := [o_1, o_2, \dots, o_T], \quad (1)$$

where each frame o_f is decomposed into two feature groups:

$$o_f := [x_f^{inv}, x_f^{env}], \quad (2)$$

where x_f^{inv} denotes collision-relevant features (e.g., the position of the delivery robot and the static obstacle) and x_f^{env} denotes collision-irrelevant environmental objects (e.g., the positions of surrounding vehicles and garbage cans).

Similarly, the action $a \in \mathcal{A}$ for each object is defined as:

$$a := [a_{inv}, a_{env}], \quad (3)$$

where a_{inv} comprises the spawn positions of collision relevant objects and the trigger position at which the pedestrian initiates an atypical behavior (e.g., falling, suddenly running), and a_{env} denotes the spawn position of environment objects.

We consider a safety-critical interaction between a delivery robot u_{robot} , controlled by a given AV algorithm, and a pedestrian u_{ped} exhibiting atypical behaviors. The pedestrian transitions from normal walking to atypical behaviors based on predefined parameters, while other objects are managed

by a traffic controller. The goal is to learn a conditional distribution $P_\pi(x | c)$, where condition c comprises collision rewards or other risk-relevant metrics.

Formally, given a training dataset $\mathcal{D}_{train} = \bigcup_{e=1}^E \mathcal{D}_e$ collected from diverse environments, our goal is to learn a conditional generative model $P_\theta(a | s, r = 1)$ that satisfies the following criteria.

- **Efficiency:** The model should generate the initial parameters of hazardous scenarios to accelerate the generation of safety-critical scenarios.
- **OOD Generalization:** The model should maintain generative performance in unseen environments $e_* \notin \{1, 2, \dots, E\}$ that are not encountered during training.

B. Invariant learning across environments

A key challenge is that the raw state s contains collision-relevant features and environment-specific features that are spuriously correlated with the reward signal. If the diffusion model conditions directly on s , it can exploit environment-specific shortcuts that fail to transfer to unseen environments. To address this, we integrate a VAE with a DANN to extract causally invariant features from multi-environment data. The resulting latent representations serve as conditioning signals for the diffusion model, ensuring robust scenario generation across unseen environments.

The VAE encodes s in a latent representation z and is trained with a reconstruction loss and a Kullback-Leibler (KL) Divergence regularization term:

$$\mathcal{L}_{enc} = \|\hat{s} - s\|_2^2 + D_{KL}(q_\phi(z | s) \| \mathcal{N}(0, I)), \quad (4)$$

where $q_\phi(z | s) = \mathcal{N}(z; \mu_\theta(s), \sigma_\theta^2(s))$ is the approximate posterior distribution over z , and $\hat{s}$ is the reconstructed state.

To prevent VAE from encoding collision-irrelevant features, we integrate DANN training with a gradient reversal layer (GRL). This method forces the encoder to discard non-causal environmental factors and yields environment-invariant latent representations. A domain classifier D_φ is trained in the latent space to predict the source environment e of z . The GRL acts as an identity function during the forward pass, but reverses the gradient by a factor of λ during backpropagation:

$$GRL_\lambda(z) = z, \quad (5)$$

$$\frac{\partial GRL_\lambda}{\partial z} = -\lambda I, \quad (6)$$

where I is the identity matrix.

The domain classification loss is the standard cross-entropy over environment labels:

$$\mathcal{L}_{domain} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(D_\varphi(z_i)_{e_i})}{\sum_{m=1}^E \exp(D_\varphi(z_i)_{e_m})}, \quad (7)$$

where B is the batch size, e_i is the ground-truth environment label of the sample i , and m indices on all training environments E .

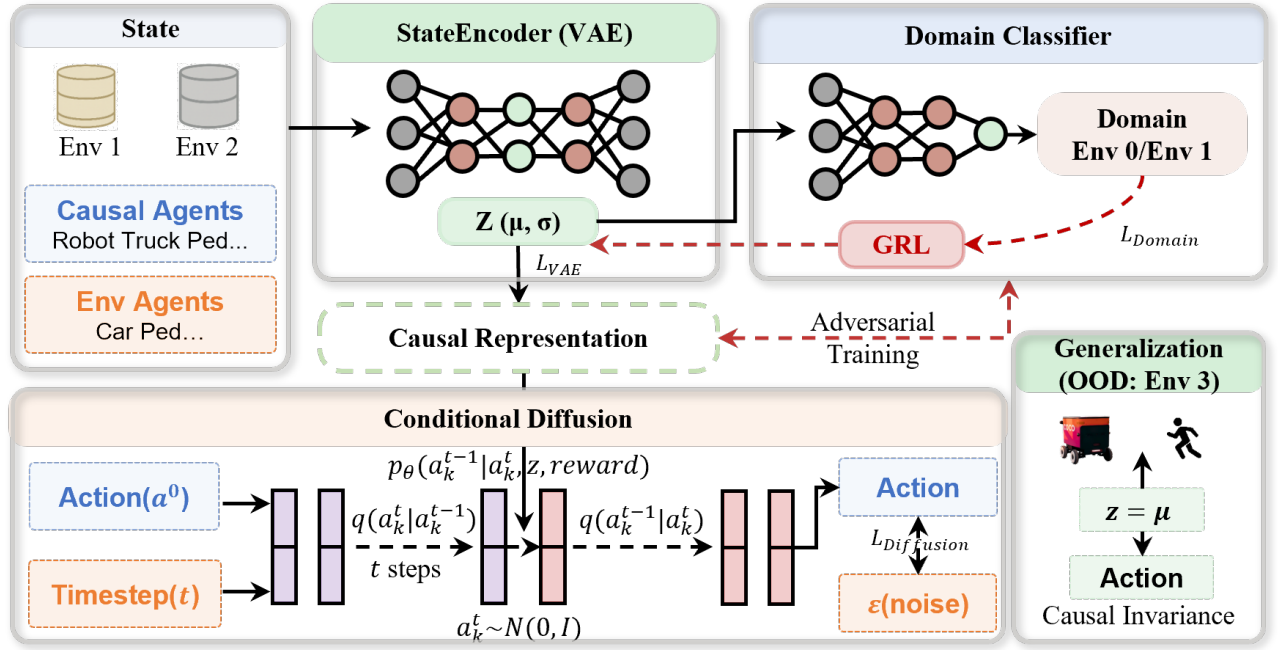


Fig. 2. Overview of Causal Decision Diffuser. Given the data from different environments, the VAE and the Domain Classifier use adversarial training to extract causal features. During the diffusion generation process, causal features and collision rewards serve as conditional signals to guide the model in generating safety-critical initial parameters. Finally, the CDD model is evaluated in unseen environments to validate its generalization ability.

This loss drives opposite optimization for the VAE and domain classifier. For D_ϕ , the gradient flows normally, training it to identify the source environment. For the VAE, the GRL reverses the gradients, penalizing environment-discriminative representations. Therefore, z becomes environment-invariant while retaining collision-relevant information.

One key design is the progressive schedule for λ :

$$\lambda(p) = \frac{2}{1 + \exp(-\gamma \cdot p)} - 1, \quad p = \frac{t_{curr}}{T_{max}} \in [0, 1], \quad (8)$$

where γ controls the steepness, p denotes the training progress, t_{curr} is the current training step and T_{max} is the total training steps.

The schedule initializes $\lambda = 0$, allowing the VAE to first learn a good representation under $\mathcal{L}_{enc}$ and following diffusion loss $\mathcal{L}_{diff}$. Then, λ gradually increases towards 1 to impose full adversarial pressure to shield non-causal information.

C. Conditional Diffusion-based Scenario Generation

We adopt the denoising diffusion probabilistic model (DDPM) to model the data distribution. The diffusion model decomposes the generation process into a sequence of denoising steps from gaussian noise, which has been shown to produce high-quality samples from complex distributions. Thus, using causal feature and collision rewards as conditions, we use the reserve diffusion process to generate safety-critical scenarios with cross-environment generalization.

Given an initial parameter of the ground-truth scenario a^0 , the forward process progressively destroys the a^0 by injecting gaussian noise in K steps:

$$q(a^k | a^{k-1}) = \mathcal{N}(a^k; \sqrt{1 - \beta_k} a^{k-1}, \beta_k I), \quad (9)$$

where $\{\beta_k\}_{k=1}^K$ follows a cosine schedule that adds to $\mathcal{A}$.

The denoising generation is an opposite process. Starting from pure noise $\mathcal{N}(0, I)$, the generative model learns to recover the initial parameters of the scenario, conditioned on the causal feature z and the collision reward r :

$$p_\theta(a^{k-1} | a^k, z, r) = \mathcal{N}(a^{k-1}; \mu_\theta(a^k, z, k, r), \tilde{\beta}_k I), \quad (10)$$

where $\tilde{\beta}_k$ is the fixed posterior variance. The distribution of the generated action is shown as:

$$p_\theta(a^0 | z, r) = p_\theta(a^K) \prod_{k=1}^K p_\theta(a^{k-1} | a^k, z, r). \quad (11)$$

During the training stage, instead of directly estimating μ_θ , we follow the ε prediction method. r will dropouts with probability p_{drop} for the following classifier-free guidance denoising process:

$$\mathcal{L}_{diff} = \mathbb{E}_{\varepsilon, k, a^0} \left[\|\varepsilon - \varepsilon_\theta(a^k, z, k, r)\|^2 \right]. \quad (12)$$

Therefore, the state encoder q_ϕ , the domain classifier D_ϕ , and the noise prediction network ε_θ are trained jointly:

$$\mathcal{L}_{total} = \mathcal{L}_{diff} + \lambda_{enc} \mathcal{L}_{enc} + \lambda_{dann} \mathcal{L}_{domain}, \quad (13)$$

where λ_{enc} and λ_{dann} are scalar weights. The total loss $\mathcal{L}_{total}$ induces three gradient pathways in the state encoder. $\mathcal{L}_{diff}$ provides a signal to make z informative for denoising,

Algorithm 1: Causal Decision Diffuser

Input: Dataset $\mathcal{D}$ from environments $\mathcal{E}$, diffusion steps K , learning rate α , guidance scale ω

// Training

while ϕ not converged **do**

 Sample $(\mathbf{s}, \mathbf{a}^0, r, e) \sim \mathcal{D}_{\text{train}}$;

// Encode causal feature by VAE and DANN

$\boldsymbol{\mu}, \boldsymbol{\sigma} \leftarrow q_\phi(\mathbf{s}); \quad \mathbf{z} \leftarrow \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \epsilon_z,$

$\epsilon_z \sim \mathcal{N}(\mathbf{0}, \mathbf{I});$

$\mathcal{L}_{\text{enc}} \leftarrow \frac{\|g_\phi(\mathbf{z}) - \mathbf{s}\|^2}{2} + D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{s})\|\mathcal{N}(\mathbf{0}, \mathbf{I}));$

$\lambda \leftarrow \frac{2}{1 + \exp(-\gamma \cdot p)} - 1;$

$\mathcal{L}_{\text{domain}} \leftarrow -\log \text{softmax}(D_\varphi(\text{GRL}_\lambda(\mathbf{z})))_e;$

// Conditional diffusion

$k \sim \mathcal{U}\{1, K\}; \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I});$

$m \sim \text{Bernoulli}(1 - p_{\text{drop}});$

$\mathbf{a}^k \leftarrow \sqrt{\bar{\alpha}_k} \mathbf{a}^0 + \sqrt{1 - \bar{\alpha}_k} \epsilon;$

$\mathcal{L}_{\text{diff}} \leftarrow \|\epsilon - \epsilon_\theta(\mathbf{a}^k, \mathbf{z}, k, m \cdot r)\|^2;$

// Parameter update

$\phi \leftarrow \phi - \alpha \nabla_\phi(\mathcal{L}_{\text{diff}} + \lambda_{\text{enc}} \mathcal{L}_{\text{enc}}) +$

$\alpha \lambda \lambda_{\text{dann}} \nabla_\phi \mathcal{L}_{\text{domain}};$

$\varphi \leftarrow \varphi - \alpha \lambda_{\text{dann}} \nabla_\varphi \mathcal{L}_{\text{domain}};$

$\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}_{\text{diff}};$

end

// Inference

$\mathbf{z} \leftarrow \boldsymbol{\mu}_\phi(\mathbf{s}); \quad \mathbf{a}^K \sim \mathcal{N}(\mathbf{0}, \mathbf{I});$

for $k = K \dots 1$ **do**

$\tilde{\epsilon} \leftarrow (1 - \omega) \cdot \epsilon_\theta(\mathbf{a}^k, \mathbf{z}, k, \emptyset) + \omega \cdot \epsilon_\theta(\mathbf{a}^k, \mathbf{z}, k, r);$

$(\boldsymbol{\mu}_{k-1}, \tilde{\beta}_{k-1}) \leftarrow \text{Denoise}(\mathbf{a}^k, \tilde{\epsilon});$

$\mathbf{a}^{k-1} \sim \mathcal{N}(\boldsymbol{\mu}_{k-1}, \tilde{\beta}_{k-1} \mathbf{I});$

end

return $\hat{\mathbf{a}}^0$

$\mathcal{L}_{\text{enc}}$ preserves the state structure, and $\mathcal{L}_{\text{domain}}$ suppresses environment-discriminative information via the GRL.

D. Classifier-Free Guidance

During inference, given a state s from an unseen environment, we first extract the invariant latent representation z . To generate safety-critical scenarios, we employ classifier-free guidance (CFG), enabled by the reward dropout strategy during training. At each denoising step k , the noise estimate $\tilde{\epsilon}$ is computed by extrapolating the conditional prediction away from the unconditional mode $\emptyset$:

$$\tilde{\epsilon} = (1 - \omega) \cdot \epsilon_\theta(\mathbf{a}^k, \mathbf{z}, k, \emptyset) + \omega \cdot \epsilon_\theta(\mathbf{a}^k, \mathbf{z}, k, r), \quad (14)$$

where $r = 1$ targets the collision and $\omega > 1$ serves as a guide scale to amplify the reward signal. The complete procedure for the proposed CDD model is summarized in Algorithm 1.

IV. EXPERIMENTS

In this section, we evaluate the proposed CDD model in three representative pre-collision traffic scenarios inspired by real-world delivery robot corner cases. Comprehensive metrics are used to quantify the efficiency of the model

in generating safety-critical scenarios and its generalization to unseen environments. Then four offline RL methods and three generative methods are used as baselines to compare with CDD and ablation studies are used to show the significance of causal feature extraction for OOD generalization.

Our key findings are summarized as follows: 1) Enhanced Generation Efficiency: Compared to baseline models, CDD shows good efficiency in generating safety-critical scenarios, achieving a higher collision rate. 2) Robust OOD Generalization: CDD shows superior generalization capabilities, validating that causal features effectively shield the model from collision-irrelevant features.

A. Experimental Design and Setting

Scenario Definition. Three cases are evaluated: pedestrian occluded by a truck (S1), pedestrian falling down (S2), and pedestrian appearing from a corner (S3). Each scenario spans three different environments. For S1, collision-relevant factors include the attributes of the delivery robot, the running pedestrian and the truck. The collision-irrelevant variables consist of the attributes of a surrounding vehicle and a pedestrian. For S2 and S3, only the delivery robot and the pedestrian are collision-relevant factors. Two surrounding vehicles and a trash can are collision-irrelevant factors. Within each scenario type, the data distribution of collision-relevant factors remains consistent across environments, whereas collision-irrelevant factors exhibit distributional variance.

Simulation platform. We employ CARLA as our simulation platform, using its physics engine for realistic collision. Standard pedestrian behaviors (e.g., sudden acceleration) are sourced from CARLA’s animations. For falling behavior, we combine WHAM [24] and CARLA-BSP [25] to extract human pose from video and map it to pedestrian assets in CARLA. To minimize noise in pose extraction, we utilized Sora to generate a high-quality video, resulting in cleaner motion data for the WHAM pipeline.

Baselines. We compare four offline RL and three generative baselines methods. Behavior cloning (BC) mimics the data distribution as a fundamental baseline [26]. Advantage-weighted regression (AWR) [27] and advantage weighted actor-critic (AWAC) [28] optimize high advantage actions via weighted regression. Implicit q-learning (IQL) [29] mitigates the distributional shift by replacing direct maximization of the q-values. For generative baselines, the conditional variational autoencoder (CVAE) achieves conditional generation by sampling data from the conditioned latent space [30]. The conditional generative adversarial network (CGAN) incorporates class labels in the generator and discriminator for controllable generation [31]. Meanwhile, conditional flow matching (CFM) uses conditional modeling by regressing the vector field of the conditional probability path [32].

Delivery Robot algorithms. We implement an intelligent driver model (IDM) for the delivery robot, equipped with three cameras to capture pedestrian atypical behaviors. The target velocities are set to 15, 10, and 7 km/h for S1, S2, and S3. Braking is triggered when any camera detects a pedestrian within 2.5 m of the robot, ensuring that the

TABLE I
PERFORMANCE COMPARISONS OF CDD AND BASELINES FOR SAFETY-CRITICAL SCENARIO GENERATION.

Scenario	Metrics	BC	AWR	AWAC	IQL	CVAE	CGAN	CFM	CDD
S1	CR↑	0.24	0.49	0.40	0.46	0.94	0.60	0.75	0.98
	CRD↓	0.79	0.20	0.20	0.61	1.00	1.00	0.08	0.02
S2	CR↑	0.35	0.67	0.93	0.84	0.86	0.97	0.81	1.00
	CRD↓	0.80	0.33	1.00	0.92	0.97	0.99	0.90	0.00
S3	CR↑	0.15	0.20	0.50	0.26	0.57	0.46	0.35	0.70
	CRD↓	1.00	1.00	0.74	1.00	1.00	1.00	0.14	0.14

delivery robot can brake safely in the absence of atypical pedestrian behaviors. We use these algorithms to collect data in CARLA and train models, and subsequently evaluate the algorithms in the generated safety-critical scenarios to judge whether the model can identify the failure modes of the delivery robot.

Data collection. To train the CDD model, we construct a dataset of pedestrian-delivery robot interactions within CARLA. The number of scenarios is determined by the dimensionality of collision-relevant objects. The S1 scenario has three collision-relevant objects and two irrelevant objects. To ensure sufficient coverage of the feature space, we collected 9,792 cases for this scenario. The S2 and S3 have two collision-relevant variables and three irrelevant variables. Due to the reduced dimensionality of the causal objects, we collected a smaller set of 3,696 and 4,284 cases for S2 and S3.

B. Experimental Metrics

We evaluate the performance of our model on three dimensions: effectiveness, generalization, and invariance of the environment. Effectiveness assesses whether the generated safety-critical scenarios can successfully induce failures in IDM policies. We use the **collision rate (CR)** as an indicator ($\mathbb{E}_{a \sim P} [\mathbb{I}(\text{collision}(a))]$) for in-distribution tests. Generalization evaluates the model’s ability to maintain high effectiveness in unseen environments. We define **collision rate decay (CRD)** calculated by the ratio of the collision rate in training environments to that in unseen environments ($1 - CR_{OOD}/CR_{Train}$). Domain invariance is used to assess the discriminative capacity of the CDD model’s domain classifier regarding causal features z . After adversarial training, the classifier should not be able to distinguish the source environment of the causal features. Therefore, **domain classification accuracy (DCA)** is used as a metric, where an accuracy close to random choice indicates superior invariance ($\mathbb{E}_z [\mathbb{I}(\hat{e} = e)]$).

C. Comparison between CDD and baseline methods

Table I summarizes the performance of the models in three scenarios. For effectiveness, the CDD model achieves better performance than offline RL baselines and classic conditional generative models, obtaining CR values of 0.98, 1.00 and 0.70 for S1-S3. These results show that CDD fulfills the requirement to generate safety-critical scenarios. Furthermore, offline RL baselines like AWR, AWAC, and IQL show weaker performance than CDD. BC provides a lower bound reflecting the dataset’s inherent collision rate.

Conditional generative models such as CVAE, CGAN, and CFM outperform offline RL baselines within in-distribution tests. For generalization, CDD is capable of maintaining high-efficiency generation in unseen environments, achieving the best CRD values in both scenarios. In contrast, the offline RL baselines, CVAE and CGAN do not shield collision-irrelevant objects. Consequently, the policies learned from these baselines lead to a performance decline when faced with unseen environments. Notably, while CFM shows competitive generalization on S1 and S3, its performance degrades significantly in S2, indicating limited consistency across diverse environments. Finally, the DCA values on three scenarios for CDD are 0.50 (Table II). This random classification accuracy confirms that the DANN module shields collision-irrelevant objects, ensuring that the causal features extracted by the VAE are indistinguishable from the domain classifier regarding their source environment.

D. Qualitative analysis

To further illustrate why incorporating spurious environmental correlations degrades generalization, we compare CDD against CFM in three scenarios (Fig. 3). In S1, CDD correctly captures the spatial relationship among the delivery robot, pedestrian, and truck, triggering the pedestrian’s running behavior at the critical moment to induce a collision. In contrast, CFM activates the running behavior prematurely, causing the pedestrian to cross the conflict point too early. In S2, CDD places the pedestrian at a closer lateral distance to the delivery robot and delays falling behavior. However, CFM generates a greater lateral distance and an earlier fall. Most notably in S3, CDD generates parameters that bring the pedestrian and the delivery robot to the corner of the wall, whereas CFM positions the pedestrian on the open road, eliminating the occlusion effect of the corner. In summary, non-causal objects degrade model performance in unseen environments, manifesting as disrupted spatial layouts and mistimed activation of pedestrian atypical behaviors. Extracting causal features mitigates these failure modes and improves OOD generalization.

E. Ablation Study

We conduct ablation experiments to identify the sources of CDD’s generalization gains. Two variants are used to compare with CDD: (i) the Decision Diffuser (DD), which lacks the invariant learning module, and (ii) a variant where the sign of the GRL in the DANN module is modified to positive value. The second setting removes adversarial training, aligning the gradient directions of the VAE and



Fig. 3. Qualitative analysis of CDD and CFM in unseen environments. Non-causal variables cause the model to produce disrupted spatial layouts and mistimed activation of pedestrian atypical behaviors.

the domain classifier. As shown in Table II, while the DD model shows high performance in generating safety-critical scenarios within training environments, its CRD value drops in unseen environments. In contrast, the CDD model maintains a robust CRD in all environments. These results show that while CDD inherits the fitting and optimization abilities of the DD model, its high generalization is attained through the extraction of causal-invariant features. Meanwhile, CDD (GRL+) maintains in-distribution collision rates comparable to those of CDD on S1 and S2, and even achieves a higher collision rate on S3. However, its CRD in unseen environments drops substantially, indicating that a positive GRL encourages the model to make the extracted features more environment-discriminative. This may benefit in-distribution optimization in certain scenarios, but at the cost of significantly degraded generalization.

TABLE II
PERFORMANCE COMPARISONS OF DD, CDD (GRL+) AND CDD.

Scenario	Metrics	DD	CDD (GRL+)	CDD
S1	CR↑	0.91	0.91	0.98
	CRD↓	0.89	0.52	0.02
	DCA	-	1.00	0.50
S2	CR↑	0.86	0.91	1.00
	CRD↓	0.20	0.15	0.00
	DCA	-	1.00	0.50
S3	CR↑	0.64	0.81	0.70
	CRD↓	1.00	1.00	0.14
	DCA	-	1.00	0.50

Furthermore, we investigate the influence of the GRL within the DANN module on the domain classifier and the latent causal features. When the GRL is positive, the DCA value reaches 1.00, indicating that the domain classifier can distinguish the source environment of the features extracted by the encoder. This implies that the latent space is entangled with collision-irrelevant information, which leads to the

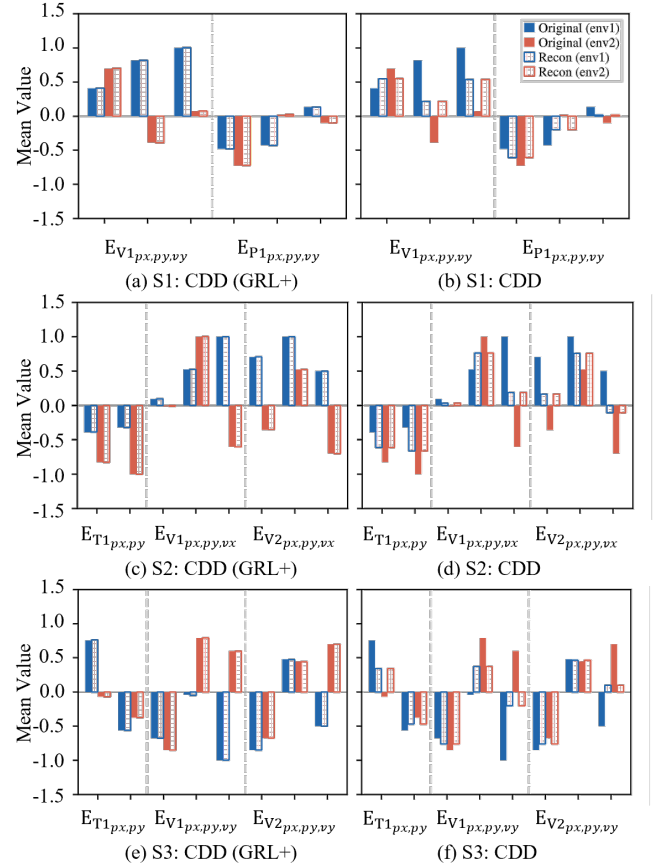


Fig. 4. Comparison of original and reconstructed states between the CDD and CDD (GRL+) models. Excluding environmental variables in CDD yields large reconstruction errors.

failure of the model in unseen environments. To further explain the role of the GRL, we visualized the normalized mean value of the original states and the reconstructed states

from the encoder for both CDD and CDD (GRL+). As shown in Fig. 4, in CDD (GRL+), the reconstructed states remain consistent with the original states, confirming that the encoder keeps the environmental difference. Conversely, when the GRL is negative (CDD), a huge discrepancy is observed between the reconstructed and original collision-irrelevant states. This suggests that the latent space successfully filters out collision-irrelevant information, achieving the extraction of causal features. Notably, this characteristic is consistently observed across three scenarios.

V. CONCLUSIONS

In this paper, we investigate the impact of atypical pedestrian behaviors on the safety of delivery robots. Based on the theory of invariant learning, we proposed a diffusion-based scenario generation model designed to enhance generalization across diverse environments. To address the scarcity of interaction data between the delivery robot and pedestrians, we developed a causal feature encoder that mitigates the influence of collision-irrelevant factors. By conditioning the diffusion model on these causal representations and collision reward, CDD can generate safety-critical scenarios across various environments. Comparisons with baseline models confirm that our approach achieves good performance on effectiveness and generalization. Future work will extend the framework beyond generating initial scenario conditions to synthesizing pedestrian pose sequences, thereby enabling closed-loop evaluation for delivery robots.

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